

Secondary references (for expansions)

The book cites a lot of literature. Here's the curated shortlist — what each is *for*, so we pull the right one when an expansion needs backup.

All of these are now **available locally** in `references/` (commercial ones supplied by you). What each is for:

On the shelf (`references/`)

- **`analytic-combinatorics.pdf`** — Flajolet & Sedgewick, *Analytic Combinatorics* (Cambridge, 2009). The authors' own advanced companion, freely distributed at `ac.cs.princeton.edu`. Our Module 5 (symbolic method) is the *appetizer*; this is the full meal. Use for: generating functions, the symbolic method, singularity analysis.
- **`TA0CP-vol1-fundamental-algorithms.pdf`** — Knuth, *Fundamental Algorithms*. Math prerequisites, sums, binomial-coefficient identities, and recurrences, done exhaustively. Good backup for the **combinatorics toolbox expansion** and Module 2.
- **`TA0CP-vol2-seminumerical-algorithms.pdf`** — Knuth, *Seminumerical Algorithms*. Random numbers, arithmetic — peripheral to us, but the definitive source when it's relevant.
- **`TA0CP-vol3-sorting-and-searching.pdf`** — Knuth, *Sorting & Searching*. ★ The crown jewel for this course: the definitive analyses of quicksort, BSTs, hashing, and more. Pull this for Modules 6, 7, 9. Dense and encyclopedic — a reference, not a read-through.
- **`TA0CP-vol4a-combinatorial-algorithms.pdf`** — Knuth, *Combinatorial Algorithms (4A)*. Deep combinatorial generation; a resource for the later application chapters.
- **`CLRS-introduction-to-algorithms-3e.pdf`** — Cormen, Leiserson, Rivest, Stein. The standard algorithms text, worst-case / design oriented (complements this book's average-case focus). The friendliest home for the **Master theorem** and a gentler second take on recurrences (Modules 2–3), plus clean pseudocode for any algorithm.
- **`sedgewick-wayne-algorithms-4e.pdf`** — Sedgewick & Wayne, *Algorithms 4e*. Same lead author; the practical/implementation companion — the actual runnable code and empirical studies behind many analyses here. Booksite: `algs4.cs.princeton.edu`. Use when you want to see an algorithm as real code + benchmarks rather than a recurrence.

- [zingaro-algorithmic-thinking.pdf](#) — Zingaro, *Algorithmic Thinking*. A gentle, problem-based intro to algorithm design (contest-flavored). Not part of the analytic track, but a friendly warm-up / sanity check when a data-structure or technique feels unfamiliar before we analyze it.

Free online, no download needed

- **Coursera — "Analysis of Algorithms" (Sedgewick)**. Video lectures tracking this exact book. Best *first pass* per chapter before reading my notes.
- [aofa.cs.princeton.edu](#) — this book's booksite: lecture slides, extra exercises, errata.
- [algs4.cs.princeton.edu](#) — code & booksite for *Algorithms* 4e.

Policy

Lazy import, same as expansions: we consult a reference only when an expansion actually needs it. The shelf is stocked; we just don't read ahead.